

FJAERLAND BREMUSEET, Norway

WMO: 013320

Lat: 61.4231N

Lon: 6.7642E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.2	-20.0	0.6	-17.8	-17.4	0.8	-14.7	8.5	4.2	7.1	3.4	1.1	10	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	24.6	16.4	22.3	15.3	20.5	14.6	17.6	22.5	16.5	20.6	15.6	19.2	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.0	19.2	14.7	10.4	18.3	13.8	9.8	17.3	49.5	22.7	46.3	20.6	43.5	19.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.1	5.0	4.1	DB	-18.6	28.2	3.8	2.3	-21.4	29.9	-23.6	31.2	-25.8	32.5	-28.5	34.2	(4)
(5)			WB	-19.9	18.8	3.5	1.1	-22.4	19.6	-24.4	20.3	-26.4	20.9	-28.9	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-2.7	-2.3	1.3	4.7	9.3	12.8	15.4	14.0	10.6	5.7	1.3	-1.4	(6)
(7)		DBStd	7.30	5.29	5.30	3.26	2.97	3.13	2.40	2.79	2.28	2.69	3.31	4.12	5.14	(7)
(8)		HDD10.0	2007	394	344	271	162	52	4	1	1	24	139	262	353	(8)
(9)		HDD18.3	4599	652	577	529	409	282	166	99	136	232	393	512	612	(9)
(10)		CDD10.0	458	0	0	0	3	29	88	167	124	42	5	0	0	(10)
(11)		CDD18.3	9	0	0	0	0	0	1	8	1	0	0	0	0	(11)
(12)		CDH23.3	121	0	0	0	0	2	28	84	7	0	0	0	0	(12)
(13)		CDH26.7	17	0	0	0	0	0	2	15	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.4	1.3	1.5	1.5	1.8	1.7	1.5	1.3	1.2	1.3	1.3	1.5	(14)
(15)	Precipitation	PrecAvg	1551	183	158	150	90	66	78	84	101	141	158	164	186	(15)
(16)		PrecMax	2281	396	359	527	154	131	208	150	188	270	391	310	393	(16)
(17)		PrecMin	842	1	4	16	40	7	18	20	23	30	34	20	15	(17)
(18)		PrecStd	315	109	99	108	32	34	47	30	43	71	90	90	110	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	9.0	9.6	18.4	23.0	26.5	28.6	24.2	20.2	17.7	12.2	9.7	(19)
(20)			MCWB	3.0	4.0	5.4	11.0	14.3	15.9	18.9	15.6	13.9	11.6	7.4	4.9	(20)
(21)		2%	DB	4.8	6.4	8.1	15.8	20.8	23.4	25.7	21.6	18.0	13.3	9.7	7.4	(21)
(22)			MCWB	1.6	2.7	4.3	9.1	12.6	14.7	17.6	15.9	13.0	9.2	6.3	3.8	(22)
(23)		5%	DB	3.3	4.7	6.6	12.9	18.1	21.1	23.1	19.9	16.4	11.6	7.9	5.2	(23)
(24)			MCWB	1.0	2.0	3.3	7.3	11.5	13.3	16.6	15.2	12.4	8.8	6.1	3.3	(24)
(25)		10%	DB	2.2	3.0	5.1	10.1	15.7	19.0	21.0	18.4	14.8	10.1	6.1	3.7	(25)
(26)			MCWB	0.7	1.7	2.5	5.8	10.0	12.5	15.9	14.6	12.2	8.1	4.7	2.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	5.0	6.6	11.7	15.0	16.6	20.2	17.9	15.3	12.4	8.6	5.7	(27)
(28)			MCDWB	5.7	8.1	8.7	17.1	21.6	23.9	26.6	21.9	17.8	15.9	10.5	8.3	(28)
(29)		2%	WB	2.4	3.4	4.6	9.4	13.4	15.2	18.3	16.7	14.3	10.7	7.4	4.4	(29)
(30)			MCDWB	3.8	5.4	7.1	14.7	18.9	21.9	23.8	20.1	16.5	12.8	9.0	6.1	(30)
(31)		5%	WB	1.6	2.4	3.7	7.8	12.1	14.1	17.3	15.9	13.3	9.4	6.0	3.6	(31)
(32)			MCDWB	2.6	4.0	6.0	12.2	16.8	19.1	22.0	18.8	15.3	10.9	7.6	5.0	(32)
(33)		10%	WB	1.0	1.7	2.9	6.4	10.8	13.3	16.3	15.0	12.4	8.4	4.8	2.7	(33)
(34)			MCDWB	1.7	2.8	4.6	9.6	15.1	17.3	20.3	17.8	14.3	9.7	6.2	3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	6.2	7.3	9.4	10.2	10.5	9.7	8.3	7.4	6.4	4.6	4.5	(35)
(36)			MCDBR	5.5	6.3	8.6	14.2	15.7	16.7	14.6	11.4	10.1	8.2	7.3	6.3	(36)
(37)			MCWBR	3.5	3.7	5.3	7.9	8.4	8.4	7.3	6.4	6.1	5.3	4.3	3.9	(37)
(38)		5% WB	MCDBR	4.7	5.4	7.5	12.5	13.9	13.5	13.6	9.7	8.8	6.8	6.4	5.5	(38)
(39)			MCWBR	3.2	3.4	4.9	7.3	7.7	7.3	7.3	5.9	5.7	4.9	4.2	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.237	0.284	0.319	0.353	0.357	0.356	0.369	0.360	0.332	0.308	0.249	0.207	(40)	
(41)		taud	2.065	2.181	2.153	2.137	2.255	2.382	2.382	2.402	2.470	2.393	2.127	1.961	(41)	
(42)		Ebn at Noon	472	684	784	829	860	871	847	823	786	664	450	319	(42)	
(43)		Edh at Noon	38	68	99	123	119	107	105	95	74	56	35	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.21	0.71	1.90	3.61	4.78	4.87	4.62	3.52	2.18	1.00	0.30	0.13	(44)	
(45)		RadStd	0.05	0.20	0.40	0.61	0.69	0.52	0.50	0.36	0.28	0.15	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page